CSE461

Introduction to Robotics Lab

Lab No. : 01
Group : 06
Section : 04
Semester : Fall_2025

Group members :

Rifat Mahmud Tamim	21201787
Faishal Monir	22101235
Rajin Ibna Rajuanur Rahman	22101717
Umma Salma Mim	22101870

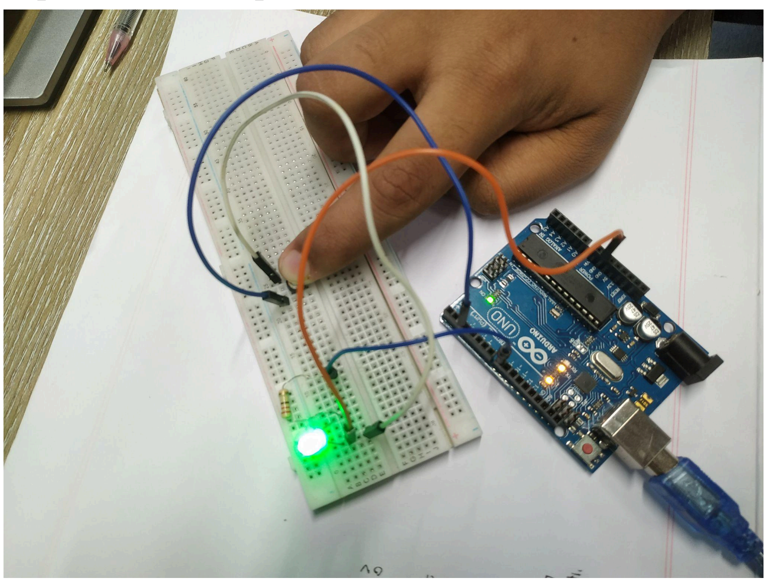
Submitted Date: 29-10-2025
Submitted to - Aabrar Islam and Purba Tahia Hassan

1. **Objectives:**

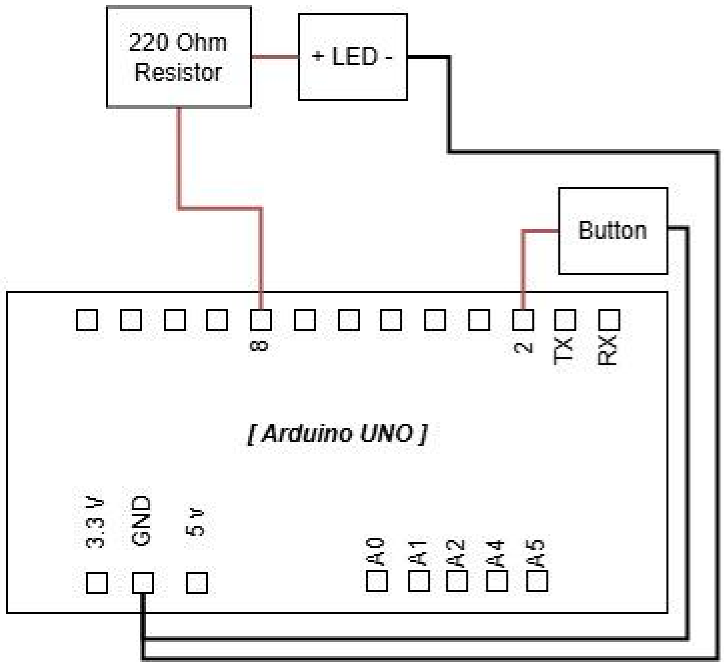
    The objective of this experiment is to create and implement a simple system that is capable of controlling a LED by using a push-button. After successfully completing this experiment, we expect to learn basic principles of digital input/output operations and get to know how to connect a switch and an LED to carry out controlled lighting functions.

2. **Equipments:**
    - Arduino Uno R3
    - LED (Light Emitting Diode)
    - Push-button switch
    - Resistor (220Ω for LED)
    - Breadboard
    - Jumper wires

3. **Experimental Setup:**



Circuit diagram:



Using the equipments first we connected the resistor with digital pin 8 from the Arduino board, then the other terminal was connected to the LEDs Positive terminal. The LEDs negative terminal was connected to the ground according to our circuit diagram. Similarly, next to the buttons, one end was connected to the Arduino digital pin 2 and the other one can connect to the common ground according to our diagram.

4. **Code:**

```
#define LED_PIN 8
#define BUTTON_PIN 2

void setup() {
      pinMode(LED_PIN, OUTPUT);
      pinMode(BUTTON_PIN, INPUT_PULLUP);
      Serial.begin(9600);
      }

void loop() {
      int buttonState = digitalRead(BUTTON_PIN);


            if (buttonState == LOW) {
                     digitalWrite(LED_PIN, HIGH);
                     Serial.println("Button pressed");
                     }

             else {
                      digitalWrite(LED_PIN, LOW);
                      Serial.println("Button released");
                         }


  delay(100);
}
```

5. **Results :**

Upon observation, the output of the experiment was the hybrid result of the previously done experiments. When we press the button, the LED lights up and in the serial monitor it prints “Button pressed”. When the Button is released, the LED turns off and the serial monitor prints “Button released”.

Button State	Output	Serial Monitor
Pressed	LED ON	Button Pressed
Released	LED OFF	Button Released

6. Discussions/Answers:

To sum up the results from this experiment, it is clear that Arduino-based microcontrollers can be used with peripherals such as buttons and sensors to view or indicate an output, in our case a LED. This is more suitable for low-cost applications or embedded systems.

But the downside is that the memory of the system is low, so scalability is not possible to a factor. During our experiment, we also encountered code compilation and uploading errors using the Arduino IDE due to connection problems. Later, it was fixed upon reconnecting the Arduino with the pc, so connection issues and memory management must be handled for large scale implementations to avoid system errors.